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Spatially Aware Peak Ranking in Mass Spectrometry Imaging

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***Abstract*—Mass spectrometry imaging (MSI) records thousands of mass-to-charge (m/z) channels at every pixel of a tissue section, so analysis usually begins by selecting a few informative peaks. Learned methods such as msiPL rank peaks from an unsupervised model, and S^{3}PL adds spatial information through 3D convolution. We ask whether local neighbourhood context improves learned peak ranking and where in the pipeline it matters. Variational autoencoders with and without neighbourhood input were fitted, and their latent spaces were segmented with a Gaussian mixture model. Peaks were ranked either by Integrated Gradients (IG) through the model or by correlating each ion image with the segment posterior maps. On 24 sections from three public datasets, at matched peak budgets and under predeclared decision rules, correlation outperformed IG in 140 of 144 paired comparisons across three seeds, a difference that was statistically supported at bin level (nominally, as sections are not independent) for every dataset and both model variants. In a one-seed check with the number of segments chosen without labels, correlation still beat IG on two of the three datasets. Neighbourhood input to the encoder gave no consistent, spatially specific gain, whereas Moran's I, a spatial statistic, was the strongest label-free ranking on two datasets and within 0.01 of the best on the third.**

***Index Terms*—mass spectrometry imaging, peak selection, variational autoencoder, spatial context, Integrated Gradients, reproducibility**

I. INTRODUCTION

Mass spectrometry imaging (MSI) records a full mass spectrum at every pixel of a tissue section, so the spatial distribution of many molecules can be studied at once without choosing targets in advance [Caprioli et al. 1997; Schwamborn and Caprioli 2010; Buchberger et al. 2018]. The resulting data are large and high-dimensional. The three collections used here, glioblastoma (GBM), colorectal adenocarcinoma (CAC) and renal cell carcinoma (RCC), contain 972 to 4,761 pixels per section and between 1,481 and 85,062 mass-to-charge (m/z) bins per spectrum. Most analyses therefore begin by selecting a small set of informative peaks.

Classical *peak picking* applies smoothing, normalisation and signal-to-noise thresholds to each spectrum on its own [Gibb and Strimmer 2012; Deininger et al. 2011]. It depends on manually set thresholds, can discard weak but informative ions [Lieb et al. 2020] and ignores *where* in the tissue a signal occurs. *Peak learning* instead treats selection as an unsupervised representation-learning problem: a model is fitted to the spectra of a section and m/z bins are ranked by how the model uses them [Abdelmoula et al. 2021]. msiPL trains a variational autoencoder (VAE) on individual spectra and reads peaks from its weights [Abdelmoula et al. 2021]. S^{3}PL adds spatial information by convolving over small tissue patches and reconstructing the whole neighbourhood, and reports large gains over msiPL [Weigand et al. 2026a].

S^{3}PL changes the architecture, the objective, the normalisation and the selection rule at the same time, so it is not clear which ingredient produces its gain. It also leaves open whether cheaper local context, such as the mean spectrum of a pixel's neighbours, can give a lightweight VAE a similar benefit, and, once a model has been fitted, which part of the pipeline actually determines peak quality. We study this with *Spatial-msiPL*, an msiPL-style VAE that can encode a spectrum together with its neighbourhood. Its latent space is segmented with a Gaussian mixture model (GMM), and peaks are ranked either by Integrated Gradients (IG) [Sundararajan et al. 2017] through the model or by correlating each ion image with the GMM posterior maps. Specifically, we ask:

RQ1 Does ranking peaks from a segmentation of the VAE latent space select better peaks than legacy msiPL's weight-based selection when both select the same number of bins?

RQ2 Which ranking mechanism on the segmentation determines peak quality, gradient attribution or direct correlation, and does the segmentation need a nonlinear encoder?

RQ3 Where does spatial information help: as input to the encoder, in the segmentation or in the ranking statistic?

RQ1 checks that the pipeline is worth studying; RQ2 and RQ3 locate the source of any gain, which a single headline score against a baseline cannot do. To make these comparisons trustworthy, we also reproduce both baselines and analyse how sensitive S^{3}PL is to the training seed.

The main ranking comparisons used matched bin counts per section and the same evaluator. Their decision rules were written down before the corresponding results existed, and the core model comparisons were repeated over three training seeds. The msiPL and S^{3}PL reproductions were first assessed

under their own protocols and evaluators, separately from the common scoring used for all other comparisons. The main findings are as follows.

- **Ranking.** With the segmentation held fixed, correlating each ion image with the GMM posterior maps outperformed IG in 140 of 144 paired comparisons across three training seeds. The difference was statistically supported at bin level (nominally) for every dataset and both VAE arms. With the number of segments chosen by the Bayesian information criterion (BIC) instead of the class count, it held on CAC and RCC but not on GBM.
- **Spatial information.** In these tests, neighbourhood input to the encoder gave no consistent, spatially specific gain, while Moran's I, a spatial statistic that uses no labels, was the strongest label-free ranking on GBM and RCC and within 0.01 of the best on CAC.
- **Encoder.** A linear five-component principal component analysis (PCA) segmentation was not reliably worse than the VAE, so the nonlinear encoder was not shown to be necessary.
- **Reproducibility.** msiPL reproduced. The published S^{3}PL glioblastoma score lay outside the distribution of ten reproduced seeds, and weight initialisation dominated the seed effect in two of three tested sections; data order also mattered in one GBM section.

The contributions of this report are:

1) a ranking ablation showing that segment-then-correlate outperforms gradient attribution in this pipeline, with paired statistics over three training seeds, together with a check of which part of that result depends on a class-informed number of segments;
2) evidence that a linear five-component PCA segmentation performs comparably to the VAE, so the nonlinear encoder was not shown to be necessary;
3) controlled tests of spatial context at the encoder, segmentation and ranking stages, with dataset-specific ranking outcomes;
4) a statistical account of the S^{3}PL reproduction gap and its seed sensitivity;
5) a matched-budget evaluation protocol with reproduced baselines and model-free references, released with the code.

Section II reviews MSI peak selection and the methods compared. Section III describes the data, models, evaluation, statistics and computation. Section IV presents the results, section V interprets them and discusses limitations, and section VI concludes.

II. Background and Related Work

A. MSI data and preprocessing

An MSI experiment rasters a tissue section and records a spectrum at each location, giving a data cube with two spatial dimensions and one m/z dimension [Caprioli et al. 1997; Buchberger et al. 2018]. Spectra vary in total intensity between pixels for technical reasons, so they are usually normalised, most often by total ion current (TIC), before comparison [Deininger et al. 2011]. Spectra may be kept as fine m/z bins or reduced to centroided peaks. Because neighbouring bins of one molecular peak are strongly correlated, any evaluation that counts bins can reward several bins of the same peak. This motivates the peak-level scoring used in this report (section III-C). Tools such as Cardinal [Bemis et al. 2015] and M^{2}aia [Cordes et al. 2021] provide standard data handling. For our work, the key consequence is that every method must receive the same input and be scored at the same resolution, or differences in preprocessing masquerade as differences in method.

B. Peak selection

Classical pipelines such as MALDIquant detect local maxima above a noise threshold in each spectrum [Gibb and Strimmer 2012]. Machine-learning peak pickers, so far mainly for nuclear magnetic resonance spectra, learn what a peak looks like from annotated or simulated spectra [Li et al. 2022]. Both operate spectrum by spectrum and do not use the spatial arrangement of the tissue. Learned representation methods instead rank m/z bins by how a model trained on the whole section uses them. This report compares such learned rankings with each other and with simple model-free rankings, under one shared evaluation.

C. Learned peak selection: msiPL and S^{3}PL

msiPL trains a fully connected VAE [Kingma and Welling 2014] on individual spectra, clusters the low-dimensional latent space and reads peaks from the network weights [Abdelmoula et al. 2021]. Its companion work, massNet, provides the glioblastoma (GBM) dataset used here [Abdelmoula et al. 2022]. S^{3}PL learns a 3D convolution over a spatial–spectral patch around each pixel, applies an attention step to the centre spectrum, reconstructs the full normalised patch and selects the bins that receive the most attention across pixels [Weigand et al. 2026a]. S^{3}PL also introduced the mask-based spatial-correlation F1 score (mSCF1) used to judge peak lists against expert annotations, and released the peak budgets and protocols for GBM, colorectal adenocarcinoma (CAC) and renal cell carcinoma (RCC). It reported 0.496, 0.601 and 0.470 mSCF1 on these collections, against 0.403, 0.431 and 0.159 for msiPL. Related representation-learning work in MSI includes deep models of 3D data [Inglese et al. 2017], low-dimensional ion-image embeddings [Guo et al. 2024] and IonMorphNet [Weigand et al. 2026b]. S^{3}PL's result motivates our spatial question. Its convolution, objective, normalisation and selector all differ from msiPL, so it is not clear which ingredient produces the gain, or whether a much smaller amount of spatial context would suffice. A related caution concerns how representations are judged: Mazarura et al. [2025] found that reconstruction error, cluster quality and downstream classification ranked autoencoders, PCA and non-negative matrix factorisation (NMF) differently on MSI data, so a better reconstruction need not mean better peaks.

D. Attribution and segment-then-correlate

Integrated Gradients attributes a model output to input features by integrating gradients along a path from a baseline input to the actual input [Sundararajan et al. 2017]. Applied

to a GMM posterior computed from a VAE latent space, it ranks m/z bins by how strongly they move a pixel towards a tissue segment, through the full nonlinear encoder. An older and simpler route is to segment the tissue first and then correlate each ion image with the segments [Alexandrov et al. 2010; Alexandrov and Kobarg 2011]. Our pipeline contains both routes on the same segmentation, which allows a direct comparison.

E. Spatial statistics and spatially aware methods

Ion images of biologically meaningful molecules tend to be spatially structured rather than noisy. Moran's I measures spatial autocorrelation [Moran 1950], and spatial autocorrelation and spatial-chaos measures have been used to judge the informativeness of MSI ion images [Alexandrov and Bartels 2013; Cassese et al. 2016]. Spatially aware segmentation smooths features using neighbourhood weights before clustering [Alexandrov and Kobarg 2011]. Graph-based models of spatial omics, such as SpaGCN [Hu et al. 2021] and STAGATE [Dong and Zhang 2022], propagate information between neighbouring locations. These are discussed as future work in section VI. Our study places spatial information at three points (the encoder input, the segmentation and the ranking statistic) and tests each one under the same evaluation.

F. Gap addressed

S^{3}PL shows that a spatial model can select better peaks than msiPL, but not *why*: spatial input, a different representation and a different ranking step all change at once. It is therefore unknown where in a learned pipeline spatial information pays off, and whether cheap neighbourhood context is enough. This report isolates those stages one at a time. Doing so requires every method to be scored on equal terms, so matched peak budgets, model-free references and repeated training runs are the means of the study rather than its subject.

III. Methodology

A. Data

Three public MSI collections were used (table I and figure 1). They are the three collections on which S^{3}PL was quantitatively evaluated [Weigand et al. 2026a], so both baselines could be reproduced under their own protocols and then compared with our methods under a common one.

GBM is the massNet patient-derived xenograft (PDX) glioblastoma collection, acquired by matrix-assisted laser desorption/ionisation (MALDI) on a 9.4 T Fourier-transform ion cyclotron resonance (FT-ICR) instrument [Abdelmoula et al. 2022]. It is available from the Metabolomics Workbench (project PR001292, study ST002045, doi:10.21228/M8Q70T). It contains eight sections from four PDX lines, two per line, with pathologist annotations of tumour and normal tissue. Sections from the same line are not independent, so GBM results are also summarised per line. **CAC** is a desorption electrospray ionisation (DESI) collection of 52 serial sections from one human colorectal adenocarcinoma [Inglese et al. 2017], available from MetaboLights (MTBLS415). We used the eight sections and three-class masks released with S^{3}PL [Weigand et al. 2026a], so CAC comparisons are within one specimen. Following S^{3}PL, the three classes are tumour, healthy tissue and background; because the expert annotations were not released, these masks are the predictions of a deep-learning model with a reported accuracy of 99.9%, as in the S^{3}PL evaluation. **RCC** is the DESI renal cell carcinoma collection of Dill et al. [2010], distributed with the Bioconductor package CardinalWorkflows [Bemis et al. 2015]. It contains eight runs, each a matched pair of cancer and normal tissue from one subject. Pixels outside the annotated tissue have no label; this unannotated region covers 45–78% of each run. The dataset's own workflow treats it as background, but this interpretation was not independently verified. We kept it as a third class in the primary analysis and report a cancer/normal-only sensitivity analysis (section IV-H). Measured coordinates and masks were audited for alignment in every section.

TABLE I
DATASETS. SECTIONS ARE THE UNITS OF ANALYSIS; THE INDEPENDENT UNITS BEHIND THEM DIFFER BY COLLECTION. CAC SECTIONS HAVE SLIGHTLY DIFFERENT m/z AXES; THE RANGE SHOWN IS FOR 40TOPL. GBM BINS ARE NOT EVENLY SPACED (MEDIAN 0.0004 DA). *RCC: CANCER, NORMAL AND THE UNANNOTATED REGION.

	GBM	CAC	RCC
Tissue	mouse PDX brain	human colon	human kidney
Ionisation	MALDI	DESI	DESI
Sections	8	8	8
Independent units	4 PDX lines	1 tumour	8 subjects
m/z bins	85,062	1,481	10,200
m/z range	102–1000	198–1061	150–1000
Mean bin width (Da)	0.011	0.58	0.083
Pixels per section	2,071–4,524	3,243–4,761	972–3,564
Classes scored	2	3	3*
Accession	ST002045	MTBLS415	CardinalWorkflows

Four sections (GBM108_positive, GBM22_2, 160TopL and 40TopL) were used for development and tuning, and the method was frozen before any of the comparisons reported here were run. RCC was not used during development and serves as a held-out confirmation collection.

B. Pipeline

Figure 2 summarises the pipeline. For each section, a model is fitted to that section's spectra without labels, peaks are ranked from the fitted model, and the selected peaks are then scored against that section's expert masks. The design is therefore unsupervised and transductive: the mask pixels are never used as training targets or ranking inputs. The annotations do, however, set two configuration values, the number of segments and the peak budget, both disclosed below. We therefore call the GMM-based rankings *unsupervised but class-informed*, and reserve *label-free* for the rankings that use neither.

Input. Let $\mathbf{s}_i \in \mathbb{R}^D_{\geq 0}$ be the spectrum at pixel i over D bins. Spectra are normalised by total ion current (TIC), $\mathbf{x}_i = \mathbf{s}_i / \sum_d s_{id}$. For the context arms, let $\mathcal{N}(i)$ be the measured pixels among the eight neighbours of i in a 3×3 window. The uniform context is the renormalised mean

$$\mathbf{c}_i = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{x}_j, \tag{1}$$

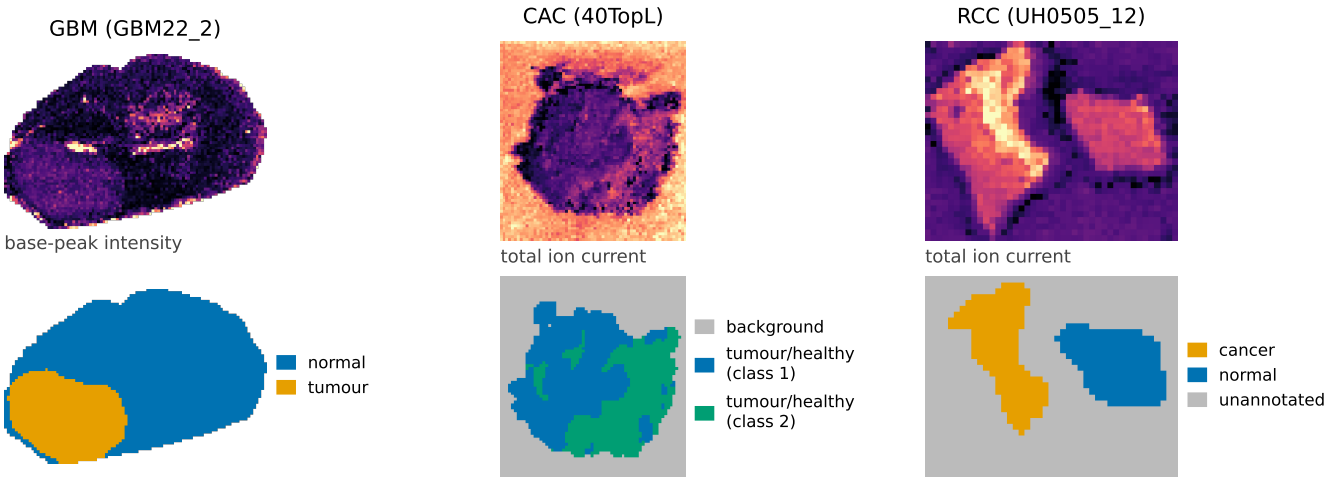


Fig. 1. One representative section per collection (GBM22_2, 40TopL and RCC_UH0505_12). Top: base-peak intensity for GBM, whose spectra are supplied TIC-normalised, and total ion current for CAC and RCC. Bottom: annotation. Unmeasured locations are blank; grey marks the CAC background and the RCC unannotated region. The two CAC tissue classes are tumour and healthy tissue, in an order the dataset does not document.

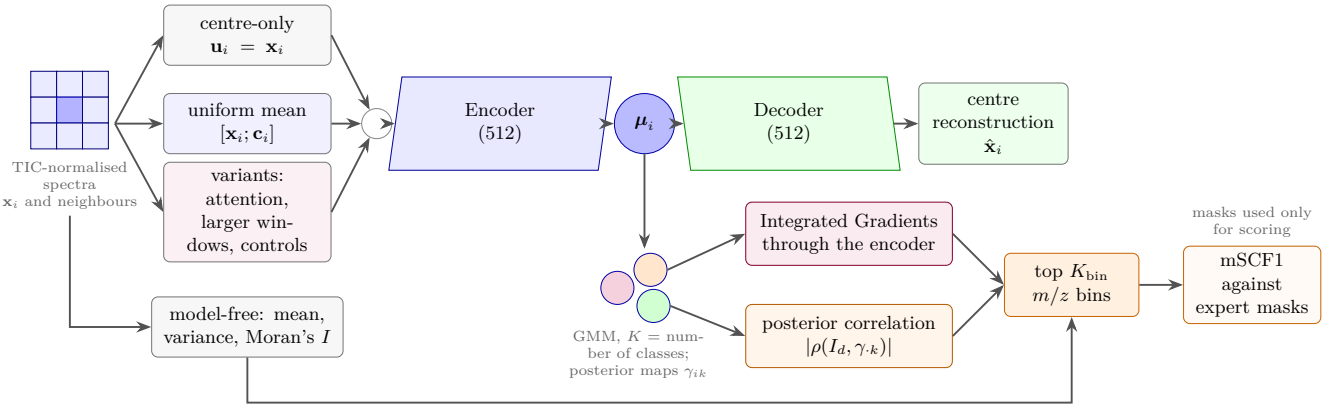


Fig. 2. Pipeline. A VAE is fitted to one section's spectra, with or without neighbourhood context; its latent means are segmented by a GMM; and m/z bins are ranked either by Integrated Gradients through the encoder or by correlation of each ion image with the GMM posterior maps. Model-free rankings use the spectra directly. Mask pixels are used only for scoring; the annotations also set the number of GMM components and the peak budget.

rescaled to sum to one, and the encoder input is $\mathbf{u}_i = \mathbf{x}_i$ (centre-only) or $\mathbf{u}_i = [\mathbf{x}_i; \mathbf{c}_i]$ (uniform). Pixels at tissue edges use only their measured neighbours, so no unmeasured location is padded; a pixel with no measured neighbour gets a zero context.

Model. Following msiPL [Abdelmoula et al. 2021], the VAE has one hidden layer of 512 units with batch normalisation (BN) and rectified linear units (ReLU) and a 5-dimensional latent space [Kingma and Welling 2014]:

$$\mathbf{h}_i = \mathrm{ReLU}\big(\mathrm{BN}(W_e \mathbf{u}_i + \mathbf{b}_e)\big), \tag{2}$$

$$\boldsymbol{\mu}_i = W_\mu \mathbf{h}_i + \mathbf{b}_\mu, \quad \log \boldsymbol{\sigma}_i^2 = W_\sigma \mathbf{h}_i + \mathbf{b}_\sigma, \tag{3}$$

$$\mathbf{z}_i = \boldsymbol{\mu}_i + \boldsymbol{\sigma}_i \odot \boldsymbol{\epsilon}, \tag{4}$$

with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I)$. The decoder mirrors the encoder and outputs $\hat{\mathbf{x}}_i = \mathrm{sigmoid}(W_o\,\mathrm{ReLU}(\mathrm{BN}(W_d \mathbf{z}_i + \mathbf{b}_d)) + \mathbf{b}_o)$, a reconstruction of the centre spectrum only. With $p_{id} = \hat{x}_{id} / \sum_{d'} \hat{x}_{id'}$, the loss is msiPL's categorical cross-entropy plus the Kullback–Leibler (KL) divergence to a standard normal prior:

$$\mathcal{L} = \frac{1}{B} \sum_{i=1}^{B} \Big[- D \sum_{d} x_{id} \log p_{id} - \tfrac{1}{2} \sum_{l=1}^{5} \big(1 + \log \sigma_{il}^2 - \mu_{il}^2 - \sigma_{il}^2\big) \Big] \tag{5}$$

over a mini-batch of B spectra. The factor D follows the msiPL implementation, which scales the cross-entropy by the number of bins so that it is on the scale of a per-bin sum; it is equivalent to weighting the KL term by $1/D$. Training uses Adam (learning rate 10^{-3}, batch size 128) for 100 epochs. For GBM ($D = 85{,}062$) the centre-only and uniform models have 87.2M and 130.8M parameters. Almost all of them are in the input and output layers: the first layer has 43.6M weights (centre-only) or 87.1M (uniform), the output layer 43.6M, and all other layers together fewer than 0.02M. Further variants replace the uniform mean with learned single-head attention over the neighbours [Vaswani et al. 2017] (about 0.5% more

parameters), use 5×5, 7×7 or 9×9 windows, replace the neighbours with zeros or with spectra shuffled from distant pixels, or hide the centre spectrum in half of the training samples so that the neighbours become necessary. Each variant was compared with the uniform arm trained under the same seed.

Segmentation. The latent means $\boldsymbol{\mu}_i$ are standardised per dimension to $\tilde{\boldsymbol{\mu}}_i$, and a Gaussian mixture model with mixing weights π_k, means $\mathbf{m}_k$ and full covariances Σ_k is fitted to them (scikit-learn, 20 initialisations, fixed seed). Its posterior

$$\gamma_{ik} = \frac{\pi_k \, \mathcal{N}(\tilde{\boldsymbol{\mu}}_i \mid \mathbf{m}_k, \Sigma_k)}{\sum_{l=1}^{K} \pi_l \, \mathcal{N}(\tilde{\boldsymbol{\mu}}_i \mid \mathbf{m}_l, \Sigma_l)} \tag{6}$$

gives, for each component k, a posterior map over the section. The number of components K equals the number of annotated classes (table I). This is a class-informed choice; the Bayesian information criterion (BIC) [Schwarz 1978], evaluated for $K = 1$ to 6, selected the largest value tested, so the main ranking comparison was repeated with BIC's choice of K (section IV-G).

Rankings. Every ranking orders all D bins, and the top K_{bin} bins are scored. The two model-based rankings use the same fitted GMM. For both, a score is computed per component and the per-component orderings are merged round-robin, skipping repeated bins, so that every component contributes equally ("balanced").

- *Integrated Gradients* attributes component k's posterior at pixel i to each input bin along a straight path from a baseline $\mathbf{u}'$, the section's mean input [Sundararajan et al. 2017], differentiating through the encoder mean $\boldsymbol{\mu}$ (no sampling of $\mathbf{z}$):

$$\mathrm{IG}_{d,k}(\mathbf{u}_i) = (u_{id} - u'_d) \int_0^1 \frac{\partial \gamma_k\big(\mathbf{u}' + \alpha(\mathbf{u}_i - \mathbf{u}')\big)}{\partial u_d} \, \mathrm{d}\alpha, \tag{7}$$

  approximated with 64 trapezoidal steps. For each component, 12 pixels are drawn uniformly at random (fixed seed) from the pixels whose GMM hard assignment is that component. IG is computed for every input dimension; for the uniform arm each bin d appears twice, once in the centre spectrum and once in the context. For bin d, the absolute attributions of its centre input and of its context input are added, $|\mathrm{IG}^{\text{centre}}_{d,k}| + |\mathrm{IG}^{\text{context}}_{d,k}|$, and bins are scored by the mean of this sum over the 12 pixels. Completeness is checked on the *signed* attributions summed over all inputs, $\sum \mathrm{IG}_{\cdot,k} \approx \gamma_k(\mathbf{u}_i) - \gamma_k(\mathbf{u}')$. A run passes if the normalised residual (the absolute gap divided by the absolute output change) has a median of at most 0.05 and a 95th percentile of at most 0.15 over its pixels. Where this check failed (4 of 144 runs), the run was recomputed from its saved model with 128 or 256 steps; in four control runs, doubling the steps changed mSCF1 by at most 0.0003.
- *Posterior correlation* scores bin d for component k by $|\rho(I_d, \gamma_{\cdot k})|$, the absolute Pearson correlation over pixels between the bin's ion image I_d and the component's posterior map $\gamma_{\cdot k}$, using TIC-normalised ion images; constant images score zero. It needs no gradients; this is the segment-then-correlate approach [Alexandrov et al. 2010].
- *Model-free rankings* use the spectra directly: mean intensity, variance and global Moran's I [Moran 1950],

$$I_d = \frac{N}{\sum_{i,j} w_{ij}} \frac{\sum_{i,j} w_{ij}(x_{id} - \bar{x}_d)(x_{jd} - \bar{x}_d)}{\sum_i (x_{id} - \bar{x}_d)^2}, \tag{8}$$

  with N the number of measured pixels and $w_{ij} = 1$ for measured Moore neighbours (the eight surrounding pixels) and 0 otherwise.
- *References:* a random ranking and a supervised oracle that ranks bins by correlation with the expert masks. The oracle is a supervised reference only.

Controls on the representation and segmentation. A *PCA control* replaces the VAE latent space with a 5-component PCA while keeping the GMM and posterior correlation, which isolates the nonlinear encoder. A *smoothed segmentation* smooths the latent means with structure-adaptive neighbourhood weights before the GMM [Alexandrov and Kobarg 2011], which places spatial information at the segmentation step instead of the encoder input.

Baselines. Legacy msiPL was reproduced from its released code, with its peak-selection threshold tuned towards the published peak counts as in the original procedure [Abdelmoula et al. 2021]. S^{3}PL was run from the public repository (v1.0.0) with the released settings: 10 epochs, Adam (learning rate 10^{-2}), batch size 16 and patch size and peaks per patch of $p = 3$, $z = 256$ (GBM), $p = 9$, $z = 256$ (CAC) and $p = 5$, $z = 1024$ (RCC) [Weigand et al. 2026a]. The released code normalises each patch by its spatial maximum, whereas the paper describes TIC normalisation, so both were tested (section IV-F).

C. Evaluation

mSCF1. For each class c and threshold $t \in \{0.3, 0.4, 0.5, 0.6\}$, the reference set R^c_t is formed as in the released S^{3}PL evaluator [Weigand et al. 2026a]: bins are sorted by their signed Pearson correlation with that class's mask, and R^c_t contains the bins ranked above the one whose correlation is closest to t. This is close to, but not exactly, the set of bins with correlation of at least t. With $R_t = \bigcup_c R^c_t$ and a selected set S,

$$\mathrm{mSCF1} = \frac{1}{4} \sum_t \frac{2\,|S \cap R_t|}{|S| + |R_t|}, \tag{9}$$

mSCF1 measures spatial agreement with expert annotation, not chemical identity.

Matched budgets. All methods select the same number of bins, K_{bin}, per section, so that differences reflect which bins are chosen and not how many. For GBM and CAC, K_{bin} is the legacy msiPL peak count with its threshold tuned to the released S^{3}PL target. On GBM these targets equal the number of bins correlating with the masks at $t = 0.4$; the paper derives the CAC targets at $t = 0.3$, but they match none of our reference sets at $t = 0.3$–0.6, so their exact derivation could not be reproduced. For RCC, K_{bin} is the released target, which equals the number of bins correlating at $t = 0.5$ to within 13 bins. Budgets of $0.5\times$ to $2 \times K_{\text{bin}}$ were checked as a sensitivity analysis. For msiPL, its saved weights were

re-thresholded over $\beta = -2.0$ to 2.5 (step 0.01) and, at each budget, the list whose length was closest to the budget was taken; every ranking was then scored at that list's own length. On RCC the longest msiPL list (1,054–1,289 bins in six of the eight runs) was shorter than K_{bin}, so those points use the longest list and are flagged; on GBM, seven points at the smallest budgets were below the shortest msiPL list and are flagged likewise, as is CAC 40TopL at 2×, where the longest list was 14% short. K_{bin} is a number of bins and is unrelated to the number of GMM components K.

Peak-level scoring (P1). Neighbouring bins of one molecular peak would otherwise each count as a true positive. P1 therefore collapses the bins of one basin of the section's mean spectrum (between two local minima) into a single peak group. Groups are split at gaps above 50 ppm, shallow basins (below 5% depth) are merged only up to a 100 ppm width, and a partition is rejected if any group spans 0.5 Da or more. A group is a reference group if its apex (highest mean-intensity bin) is in the bin-level reference set, a ranking selects a group if it selects any of its bins, and each ranking is read down until it covers K_{peak} distinct groups, the number of reference groups at $t = 0.4$. Partitions were audited and approved before any score was computed. On CAC and RCC, P1 groups are singletons or near-singletons, so P1 acts there as a second budget check rather than independent evidence.

D. Experimental design

Each comparison in section IV follows a rule written down before its outputs existed, in an append-only protocol that also records deviations and execution logs. The rules differ by comparison. In the context comparisons, a method *helps* if its mean gain is at least $+0.02$ mSCF1 and it gains in at least 6 of 8 sections; methods are *equivalent* if their mean difference lies within ± 0.02. Other complete outcomes are *mixed*. The IG-versus-correlation replication rule instead asks whether IG is lower in at least 7 of 8 sections, with a mean difference of at most -0.02, in every training run, at bin level and, on GBM and CAC, also at peak level; its predeclared labels and fall-through cases are reported as written. The threshold of 0.02 was chosen as the smallest difference we would act on; it is of the same order as the seed-to-seed standard deviation (SD) of collection means (0.002–0.026). The two core arms were trained with three seeds on all 24 sections; the variants were screened against the uniform arm with one seed each. Seed results are reported as the mean ± SD across seeds of the collection means.

E. Statistical analysis

The outcome labels are accompanied by paired non-parametric statistics over sections: the exact two-sided Wilcoxon signed-rank test [Wilcoxon 1945], the Hodges–Lehmann (HL) median paired difference with its exact 95% confidence interval (CI) [Hodges and Lehmann 1963], the matched-pairs rank-biserial correlation r and the sign count. Every pair of methods is scored on the same sections, so the data are paired and the signed-rank test is used rather than the unpaired Mann–Whitney test, which would ignore that pairing. Where three seeds exist, seeds are averaged within each section first, so the unit is the section rather than the run.

The 38 rule-labelled comparisons were grouped into five families by question: ranking (6 tests), versus legacy msiPL (4), encoder context and variants (17), segmentation (9) and S^{3}PL (2). Holm correction [Holm 1979] was applied within each family. With $n = 8$ the smallest attainable exact p-value is $2/2^8 = 0.0078$, so Holm significance is attainable only in families of at most six tests, and effect sizes and intervals carry the main weight. A comparison is called *statistically supported* only if its Holm-adjusted $p < 0.05$ *and* its outcome label points the same way. GBM is also summarised per PDX line ($n = 4$), outside the Holm families. Mixed-effects models with random intercepts for section and seed were also fitted, but for IG against correlation none converged cleanly, so they are not used. Sections are not fully independent: CAC sections are serial sections of one tumour and GBM sections come in pairs from four PDX lines. The section-level p-values and intervals are therefore nominal, and the per-line GBM summary is the more conservative check.

F. Implementation and computation

The pipeline was implemented in Python with PyTorch and scikit-learn [Pedregosa et al. 2011]. Training ran on the University's MSCluster under Slurm, on RTX 3060 and RTX 3090 GPUs (all timed training on RTX 3090), and ranking and scoring ran on CPU nodes. The code, protocol, result summaries and figure scripts are available at https://github.com/2653934/Spatial-msiPL (release `v1.1`, commit `76aa494`).

Caching. The original data loader re-read each batch from the HDF5 data file on shared storage, and a 100-epoch run on GBM108_positive (the smallest GBM section, uniform arm) took 11.3 h. Profiling showed that preparing a batch took about 24.5 s, while the forward and backward passes took about 0.02 s. Reading each section into host memory once (about 13 s) reduced the run to 23 min, a 30× speed-up, with identical loss histories and final weights. The cache costs one float32 copy of the section, $N \times D \times 4$ bytes: 0.70 GB for GBM108_positive and at most 1.54 GB for the largest GBM section, with GPU memory unchanged. The likely cause of the slowdown is many small reads over network storage, which was not measured separately.

Reproducibility checks. Before saved runs were reused, same-seed reruns were checked for exact equality of losses, scores and selected bins. Exact equality of rankings depends on the CPU model, because floating-point summation order differs between processors, so each ranking was timed on the same processor type that produced it. Saved result summaries record evaluator and protocol versions, parameters, input hashes and code hashes, while scripts refuse stale or existing outputs; exact reproduction remains hardware- and path-sensitive.

IV. Results

Scores are mSCF1 at the matched budget K_{bin}, at bin level, with differences paired by section. The two core VAE arms are given as mean ± SD over three training seeds, one-seed results are marked as such, and the statistics of section III-E are reported beside the outcome labels.

A. Baseline reproductions

Each baseline was first run under its own protocol and evaluator. msiPL reproduced closely on all three collections: 0.414 against 0.403 published on GBM, 0.402 against 0.431 on CAC and 0.157 against 0.159 on RCC. With S^{3}PL's own evaluator, the CAC mean was 0.547 $\pm$ 0.038 over ten seeds against 0.601 published. The published value lies just outside our seed distribution, and the predeclared conclusion is "not explained by seed selection alone". RCC reached 0.41–0.44 over three seeds against 0.47 published, while GBM reached 0.319 $\pm$ 0.042 against 0.496 (section IV-F).

B. Ranking by correlation outperforms IG (RQ1, RQ2)

Table II scores every ranking on the same segmentation and budget.

Against legacy msiPL (RQ1). Both VAE–IG arms beat reproduced msiPL in every GBM and CAC section: 0.518 and 0.514 against 0.414 on GBM, and 0.530 and 0.553 against 0.402 on CAC (table II). The predeclared tests, run on one seed, are statistically supported in all four comparisons, and the seed-averaged values agree (HL $+0.10$ to $+0.15$, $r = 1$, Holm $p = 0.031$). Repeating the comparison with msiPL re-tuned to $0.5\times$–$2 \times K_{\text{bin}}$, IG stayed ahead in all eight GBM sections at every budget. On CAC it stayed ahead from K_{bin} upwards, but at half the budget msiPL was about as good (IG ahead in 5/8 sections), so the predeclared requirement that the advantage hold at every budget was met on GBM but not on CAC. On RCC, where msiPL cannot produce lists as long as K_{bin} in six of eight runs, IG beat msiPL in all eight runs when both were scored at msiPL's own list length (0.37 against 0.16). This compares complete pipelines and does not show which ingredient causes the gain.

IG against posterior correlation (RQ2). Holding each fitted GMM fixed, ranking bins by their correlation with the posterior maps outperformed IG in 140 of 144 comparisons (8 sections $\times$ 3 collections $\times$ 2 arms $\times$ 3 seeds; table III and figure 3). The four exceptions were small, at most $+0.017$. The predeclared rule asked whether IG was below correlation in at least 7 of 8 sections, with a mean difference of at most -0.02, in every training run, at each applicable scoring level. It held in 16 of 18 runs; it failed for one CAC training seed in both arms (IG below in 6 of 8 sections at peak level for the centre-only arm, and at both levels for the uniform arm), so the rule's verdict is that the result "does not hold overall". The direction itself never reversed: all 30 collection means of IG minus correlation (18 at bin level and 12 at peak level) were negative.

Statistics. Separately from that replication rule, with seeds averaged within section, the difference is statistically supported (Holm $p < 0.05$ with the same direction as the outcome) in all six collection–arm comparisons at bin level (table III). These p-values are nominal, because CAC sections share one tumour and GBM sections share PDX lines; the GBM result also holds in each of the four lines. Correlation scored higher in every section (rank-biserial $r = -1$), and every raw p-value is the smallest attainable for $n = 8$, so this is as strong as eight sections can show. The seed changed the size of the gap only slightly: the three seed means of a collection differed by at most 0.05. At peak level (P1) the result is also supported, except for the CAC uniform arm (-0.054, 7/8, Holm $p = 0.078$). GBM is negative in every PDX line. On the two development sections, sampling 48 or 192 pixels per component for IG instead of 12 raised its score by at most 0.024, far less than the gap.

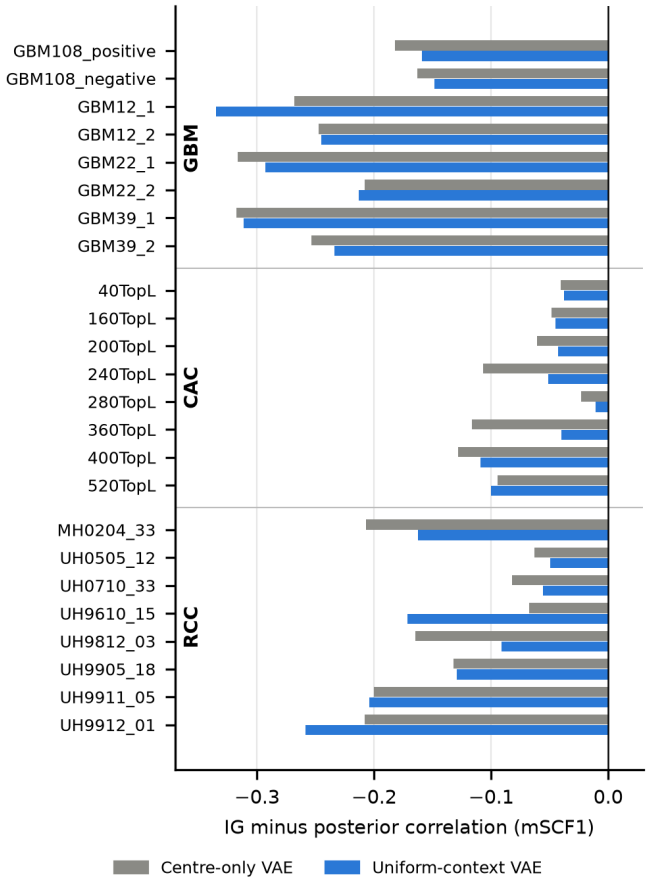


Fig. 3. IG minus posterior correlation (mSCF1) per section for both VAE arms, averaged over three seeds. Every section is negative: correlation with the segmentation outperforms IG on the same segmentation.

Answer to RQ1 and RQ2. The pipeline beats legacy msiPL, but the attribution method is not what carries that gain. Given the same segmentation, a direct correlation ranking is better than IG on all three collections when K matches the annotated classes, and on CAC and RCC also when K is chosen without labels (section IV-G). The useful signal is in the segmentation, not in the gradients, and on GBM it depends on that segmentation matching the tissue classes.

C. Simple rankings and the metric

Among the unsupervised rankings, the strongest differs by collection (table II): posterior correlation on GBM (0.762, with Moran's I at 0.684), mean intensity on CAC (0.612, with correlation, Moran's I and variance all within 0.02) and Moran's I on RCC (0.631, above correlation at 0.54–0.56). CAC is close to saturation at its budget: all methods, including the oracle, converge near 0.41 at twice the budget, which limits how much CAC differences can reveal. On GBM and

TABLE II

COLLECTION MEAN MSCF1 AT K_{BIN} (BIN LEVEL). IG AND POSTERIOR CORRELATION USE THE SAME FITTED GMM; MODEL-FREE RANKINGS USE THE SPECTRA. BOLD: BEST UNSUPERVISED RANKING PER ROW. THE ORACLE USES THE EXPERT MASKS AND IS A SUPERVISED REFERENCE. IG AND POSTERIOR CORRELATION: MEAN ± SD OVER THREE TRAINING SEEDS; THE OTHER RANKINGS DO NOT DEPEND ON THE SEED.

Collection	VAE arm	IG	Posterior corr.	Moran's I	Mean int.	Variance	Legacy msiPL	Random	Oracle
GBM	centre	0.518 ± 0.006	**0.762 ± 0.021**	0.684	0.483	0.497	0.414	0.009	0.809
GBM	uniform	0.514 ± 0.006	**0.757 ± 0.017**	0.684	0.483	0.497	0.414	0.009	0.809
CAC	centre	0.530 ± 0.007	0.608 ± 0.004	0.605	**0.612**	0.593	0.402	0.108	0.654
CAC	uniform	0.553 ± 0.006	0.608 ± 0.002	0.605	**0.612**	0.593	0.402	0.108	0.654
RCC	centre	0.403 ± 0.013	0.543 ± 0.025	**0.631**	0.498	0.508	0.157[a]	0.171	0.756
RCC	uniform	0.421 ± 0.011	0.561 ± 0.019	**0.631**	0.498	0.508	0.157[a]	0.171	0.756

[a] Scored at msiPL's own list length (median 1,158 bins against a median K_{bin} of 1,688), because msiPL cannot reach K_{bin} in six of eight runs. At that length (means over the eight runs, one training seed) IG scores 0.37 and posterior correlation 0.49.

TABLE III

IG MINUS POSTERIOR CORRELATION (BIN LEVEL). MEAN ± SD ACROSS THREE TRAINING SEEDS OF THE COLLECTION MEANS; IG LOWER: SECTION COMPARISONS IN WHICH IG SCORED LOWER, OVER 8 SECTIONS × 3 SEEDS; HL: HODGES–LEHMANN MEDIAN DIFFERENCE OVER SECTIONS (SEEDS AVERAGED) WITH EXACT 95% CI; HOLM ADJUSTS WITHIN THE SIX-TEST RANKING FAMILY.

	IG − corr.	IG lower	HL [95% CI]	Holm p
GBM centre	-0.244 ± 0.018	24/24	-0.248 [-0.293, -0.186]	0.047
GBM uniform	-0.242 ± 0.011	24/24	-0.241 [-0.311, -0.181]	0.047
CAC centre	-0.077 ± 0.003	23/24	-0.078 [-0.116, -0.040]	0.047
CAC uniform	-0.055 ± 0.007	22/24	-0.046 [-0.080, -0.027]	0.047
RCC centre	-0.141 ± 0.014	23/24	-0.137 [-0.204, -0.073]	0.047
RCC uniform	-0.140 ± 0.026	24/24	-0.139 [-0.210, -0.070]	0.047
Total		140/144		

RCC, sections whose GMM agrees better with the masks have correlation scores closer to the oracle (Spearman $\rho = -0.90$ and -0.84 over the 16 section–arms of one training seed, between balanced accuracy and the gap to the oracle). mSCF1 rewards ion images that follow the expert regions, so a segmentation that resembles those regions scores well when correlated against; this alignment is part of why correlation does well (section V).

D. Spatial context in the encoder and the segmentation (RQ3)

- **Uniform 3×3 context** helps IG on CAC ($+0.023$ at bin level and $+0.027$ at peak level over three seeds, gaining in 8/8 sections; per seed: helps, mixed, equivalent), although this is not statistically supported (Holm $p = 0.13$), and it gives no gain under correlation ranking (0.000). On GBM and RCC, context and centre-only are equivalent (-0.004, 3/8 and $+0.018$, 6/8). Across the four GBM PDX lines, context is slightly positive in one line ($+0.016$) and negative in three.
- **Spatial specificity.** If the CAC gain came from correct local placement, real neighbours should beat neighbours shuffled from distant pixels. They tied (0.542 against 0.543). On GBM, zeroed context scored highest (0.459, against 0.446 for real). Both checks used IG, one training seed and the seven confirmation sections of each collection.
- **Larger windows** did not help: 7×7 and 9×9 were 0.035 below 3×3 on GBM (mixed; higher in only 2/8 sections) and equivalent on CAC (at peak level -0.018, higher in 1/8 sections for 7×7; figure 4).
- **Learned attention** was equivalent to uniform averaging on both collections ($+0.011$ GBM, -0.006 CAC). With real neighbours its weights stayed almost uniform (entropy divided by its maximum, log of the number of neighbours: 0.98–0.999). With shuffled neighbours it learned selective weights, so the mechanism works; real neighbours are simply near-copies of the centre (mean cosine similarity between centre and neighbour spectra 0.97–0.98). At tissue boundaries, where neighbours differ most, attention did depart further from uniform on the two development sections (median half-L_1 distance from equal weights 0.011–0.028 against 0.003–0.011 in the interior), but it favoured neighbours of the same tissue class in only 50–56% of boundary pixels on CAC and 41–50% on GBM, so it did not learn to follow tissue boundaries. Masking the centre during training did not make attention selective, and it lowered GBM scores by 0.039. On the two development sections, replacing learned attention weights with equal weights at inference slightly raised peak scores ($+0.0099$ and $+0.0038$); this same-checkpoint diagnostic did not retrain a uniform model.
- **Spatially smoothed segmentation** never helped: four comparisons were equivalent and two were mixed, GBM uniform ($+0.029$, driven by two sections) and RCC uniform (-0.021, 4/8).

Answer to RQ3. Neighbourhood information entering the encoder or the segmentation gave no consistent, spatially specific gain in this pipeline. The spatial information that did help sits in the ranking statistic: correlation with spatially coherent segment maps, and Moran's I, which ranks bins by how spatially structured their ion images are.

E. Is the nonlinear encoder needed? (RQ2)

Replacing the VAE latent space with a 5-component PCA, while keeping the GMM and correlation ranking, gave VAE − PCA-5 differences, averaged over three seeds, of $+0.042$ on GBM (higher in 5/8 sections), -0.012 on CAC and -0.010 on RCC. The predeclared outcomes were mixed on GBM and CAC (on CAC because the peak-level difference was -0.028) and equivalent on RCC, and seed averaging did not change them. VAE segmentations matched the masks slightly better (balanced accuracy 0.868 against 0.839 on GBM) without reliably better peaks. Reconstruction did not separate the two either: PCA-5, fitted once per section, had *lower* reconstruction

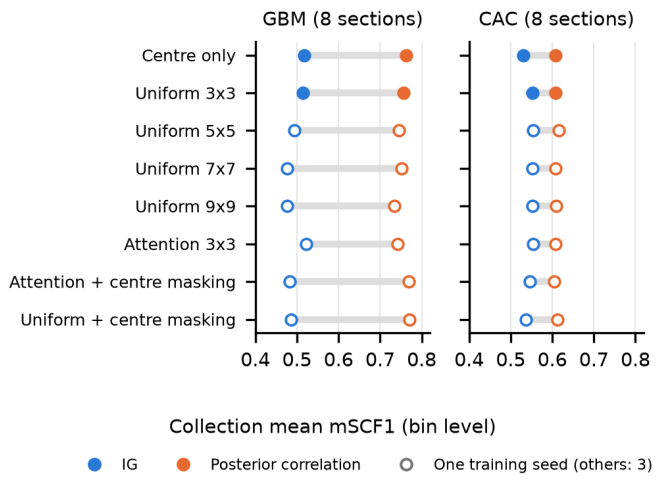


Fig. 4. Collection-mean mSCF1 (bin level) of IG and posterior correlation for each neighbourhood variant. Filled markers: three training seeds; open: one seed. No variant gives a consistent, spatially specific gain.

error (mean squared error on TIC-normalised spectra) than the one-seed centre-only VAE on all eight GBM sections, and slightly higher error on 7/8 CAC sections. The choice of the number of GMM components is examined in section IV-G.

Training length. On two development sections (GBM108_positive and 40TopL), both core arms were also scored after 10, 25, 50 and 100 epochs. The training loss fell mostly by epoch 25 and changed little afterwards (on GBM, from 593,007 to 575,894 and then to 571,502 at epoch 100). mSCF1 did not improve with longer training: across the four checkpoints IG varied by at most 0.04 and posterior correlation by at most 0.03, with no consistent trend. Epoch 100 reproduced the production scores and rankings exactly, so the 100-epoch setting was not chosen on the score.

F. S³PL: neighbour dependence, reproduction and seeds

Neighbour dependence. Retraining S³PL with each patch filled by copies of the centre spectrum collapsed it on CAC from 0.591 to 0.15–0.18 (8/8 sections). Neighbours shuffled from distant pixels recovered most of the score (0.55), and correct placement added about $+0.04$, but inconsistently (mixed, Holm $p = 0.63$). S³PL therefore needs neighbour input under its patch objective, but a consistent benefit from *correctly placed* neighbours was not established. Training for longer made it worse (CAC 0.591 to 0.336 and GBM 0.316 to 0.212 from 10 to 100 epochs; epoch 10 was best in 15 of 16 sections). The released 10-epoch protocol therefore acts as an effective early-stopping choice, but these scores provide no label-free rule for selecting that checkpoint.

Reproduction gap and seeds. Over ten released seeds, scored with S³PL's own evaluator, the published GBM score of 0.496 lies above all ten seed means (maximum 0.374) and above every seed in 6/8 sections; even the best of ten seeds in every section reaches only 0.434. On CAC, the best of ten reaches 0.600, just below the published 0.601. Replacing the released spatial-maximum normalisation with the TIC normalisation described in the paper did not close the GBM gap (0.303 against 0.324). We therefore leave the GBM gap unexplained; untested candidates are section-specific data preparation and hyperparameters tuned on the evaluation sections. The seed effect itself is specific: about half of the score variance lies between sections, almost none in a seed main effect and about 47% in the residual (section × seed interaction plus run noise). Seeds switch particular sections between distinct solutions (GBM108_positive between 0.02 and 0.46). On three sections that switched in this way, weight initialisation dominated in two (SD 0.222 against 0.001 from data order on GBM108_positive), while data order also mattered on GBM108_negative. The measured contribution of GPU nondeterminism was limited in the tested repeats.

Summary. msiPL reproduced. S³PL's own results were close on RCC and CAC, with the published CAC value just outside our seed distribution, but its published GBM value remains unreproduced. Individual sections can switch between distinct solutions, making single-run comparisons unreliable.

TABLE IV
CORRELATION MINUS IG (MSCF1, BIN LEVEL, ONE SEED) WITH K EQUAL TO THE CLASS COUNT AND WITH BIC'S $K = 6$. CORR. HIGHER: SECTIONS IN WHICH CORRELATION WAS HIGHER AT $K = 6$. LAST COLUMN: THE STRONGEST UNSUPERVISED RANKING AT $K = 6$.

	Class K	$K = 6$	Corr. higher	Best at $K = 6$
GBM centre	$+0.225$	-0.019	2/8	Moran's I (0.684)
GBM uniform	$+0.233$	$+0.006$	5/8	Moran's I (0.684)
CAC centre	$+0.078$	$+0.045$	8/8	mean int. (0.612)
CAC uniform	$+0.047$	$+0.025$	7/8	mean int. (0.612)
RCC centre	$+0.156$	$+0.073$	8/8	Moran's I (0.631)
RCC uniform	$+0.169$	$+0.054$	7/8	Moran's I (0.631)

G. Number of segments chosen without labels

Because the GMM's K equals the number of annotated classes, we repeated the ranking comparison with K chosen without labels. BIC selected $K = 6$ in all 48 section–arms, the largest value tested ($K \leq 6$), so this is BIC's choice at the boundary of the range rather than a clear optimum. Using the existing models (one seed), we refitted the GMM at $K = 6$ and recomputed IG and posterior correlation; all IG runs passed the completeness check (one at 256 steps), and the rankings that do not depend on K reproduced their earlier scores exactly. The predeclared rule asked whether correlation still beat IG in at least 7 of 8 sections (table IV).

The result held on CAC and RCC in both arms but not on GBM, where correlation and IG became about equal. The change came almost entirely from correlation: with $K = 6$ it fell by 0.25–0.28 on GBM, 0.06–0.10 on RCC and 0.02–0.03 on CAC, while IG changed by at most 0.04. On GBM the six-component segmentation agrees less with the masks (adjusted Rand index 0.19 on average). In the example of figure 5, one component still follows the tumour, but the normal tissue is split into several patchy components; because every component contributes equally to the balanced ranking, a likely reason for the drop is that much of the list then follows these patches rather than the annotated regions.

IG still beat legacy msiPL in all eight GBM and CAC sections at $K = 6$. Moran's I, which uses no labels at all, was

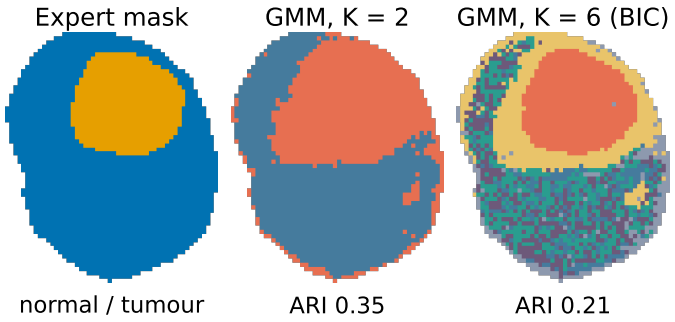


Fig. 5. GMM segmentations of GBM section GBM39_2 (uniform arm, one training seed) beside the expert mask, with each segmentation's adjusted Rand index (ARI) with the mask. The section was chosen by a rule fixed before plotting: the fourth of the eight GBM sections ordered by IG minus correlation at the class K. At BIC's $K = 6$ one component still follows the tumour, while the normal tissue is split into several components.

the strongest label-free ranking on GBM and RCC and within 0.01 of the best on CAC at either K.

H. RCC: held-out confirmation

On the held-out RCC collection, correlation outperformed IG in 23 of 24 centre-only and 24 of 24 uniform comparisons over three seeds, and Moran's I was the strongest unsupervised ranking (0.63). The primary GMM used class-informed $K = 3$. The unannotated class covers 45–78% of pixels; the Cardinal workflow treats it as background, but we did not verify that interpretation independently. The documentation describes heterogeneous tumour with large normal-looking areas in UH0505_12 and UH9905_18. When only cancer and normal pixels are scored (one training seed), which changes the question to separating cancer from normal tissue only, the advantage shrinks: correlation beats IG in 5/8 runs for the centre-only arm (and falls below mean intensity, variance and Moran's I) and in 6/8 runs for the uniform arm. The three-class RCC result may therefore partly reflect separating annotated tissue from the unannotated region; RCC is held-out confirmation of the ranking contrast under these qualifications.

I. Runtime

Table V gives median times per section. Once a segmentation exists, correlation ranking takes about a second or less, while the IG step takes seconds on CAC and minutes on GBM and RCC. The two differ in scope: correlation starts from loaded spectra and a fitted GMM, whereas the IG step also fits the GMM and runs completeness checks. Every timed ranking reproduced its saved ranking exactly. The 100-epoch VAE trains for longer than S^{3}PL's 10-epoch protocol, but over a complete workflow on CAC section 160TopL (training, selection and scoring, three repeats; figure 6) the VAE took 122–135 s against 151–174 s for S^{3}PL, with 1.2 against 1.3 GiB of host memory and 237 against 315 MiB of GPU memory. These are measurements under each method's own protocol, not an equal-work comparison.

TABLE V
MEDIAN RUNTIME PER SECTION. TRAINING ON AN NVIDIA RTX 3090 (VAE 100 EPOCHS WITH CACHED DATA, S^{3}PL 10 EPOCHS). RANKING ON CPU WITH TWO THREADS, GIVEN A TRAINED MODEL (UNIFORM ARM, ONE SEED): AN INTEL CORE I5-14400 FOR GBM AND CAC AND AN INTEL CORE I9-10940X FOR RCC. CORRELATION INCLUDES THE GMM POSTERIOR CALCULATION; THE IG STEP ALSO FITS THE GMM AND RUNS COMPLETENESS CHECKS.

	GBM	CAC	RCC
VAE training (min)	32–35	1.3–1.4	2.2
S^{3}PL training (min)	13.7	0.9	1.6
Posterior corr. (s)	1.12	0.027	0.11
Moran's I (s)	8.4	0.18	0.75
Mean intensity (s)	0.10	0.003	0.013
IG step (s)	770	6.0	372

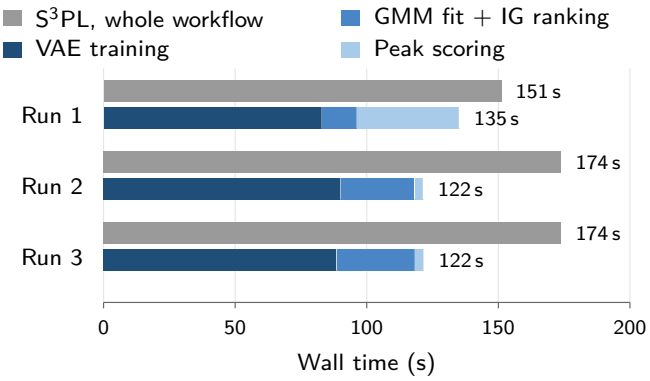


Fig. 6. Wall time of the complete workflow on CAC section 160TopL over three repeats. S^{3}PL is one bar; the VAE workflow is split into training, GMM fitting with IG ranking, and peak scoring. Each method follows its own protocol (S^{3}PL 10 epochs, VAE 100 epochs), so this is not an equal-work comparison. Peak scoring was slower in the first repeat (39 s against 3 s).

V. DISCUSSION

A. Why correlation outperforms IG here

Three explanations fit the ranking result.

1) **Different questions.** IG asks which bins would change a few sampled pixels' component probabilities. Correlation asks which ion images follow a component across the whole section, which is closer to what mSCF1 rewards.
2) **Sampling.** IG averages over 12 sampled pixels per component, while correlation uses every pixel. More pixels improved IG only slightly ($\leq +0.024$) on the two sections tested, so sampling explains little of a gap of 0.05–0.25, but other IG settings (baselines, path counts, signed attributions) remain untested.
3) **Metric alignment.** A segmentation that resembles the masks produces posterior maps that a mask-based metric naturally rewards. The label-free check supports this: with $K = 6$ correlation lost most of its advantage on GBM, where the segmentation no longer matched the two tissue classes, but kept it on CAC and RCC.

Correlation is therefore best read as an ablation finding about this pipeline: the unsupervised segmentation carries the useful signal, and gradient attribution through a dense VAE does not add to it. We do not claim that IG is generally

inferior, nor that segment-then-correlate, which is established practice [Alexandrov et al. 2010], is a new method.

B. What remains of the spatial hypothesis

The original hypothesis was that neighbourhood input would improve a VAE's representation and therefore its peaks. In a dense VAE that averages or attends over neighbours, no intervention gave a consistent gain beyond what nonlocal (shuffled) context also provides, and smoothing the segmentation did not help either. The attention experiments explain why. At 3×3 scale, adjacent spectra are near-copies of the centre, so there is little to select between. Spatially aware ranking statistics were strongest on GBM and RCC and within 0.01 of the best on CAC. S^{3}PL collapses without neighbour input under its patch objective, although correctly placed neighbours added only an inconsistent $+0.04$ over shuffled ones. In this pipeline, then, spatial information paid off when it was used to *rank* peaks, not when it was fed to the encoder: spatially aware ranking rather than spatially aware encoding is what the evidence supports, within the tested models and datasets.

C. Reconstruction is not ranking quality

PCA-5 reconstructed GBM better than the VAE yet was not reliably worse at ranking peaks, while the VAE segmented slightly better. Reconstruction error rewards fitting all intensity variation, including intense ions unrelated to tissue classes, whereas peak quality here depends on how well a low-dimensional segmentation separates the classes. This agrees with Mazarura et al. [2025].

On the two development sections, label-trained probes using within-section pixel folds recovered expert-class information from frozen latents even when the unsupervised GMM partitioned them differently; the probes are diagnostics, not label-free validation. On one section, a tied-covariance GMM agreed better with the masks but had worse BIC, so selecting it for mask agreement would be label-guided.

D. S^{3}PL instability and the reproduction gap

The S^{3}PL seed analysis shows that comparisons with a single S^{3}PL run can be misleading. Specific sections switch between distinct solutions depending on the initial weights. With TIC normalisation instead of the released spatial-maximum normalisation, this instability largely disappeared (GBM, three seeds: median per-section seed SD 0.011 against 0.061), along with the highest-scoring outcomes, so it is associated with the released preprocessing. Neither seeds nor normalisation explain the published GBM score. We therefore report the gap as unresolved, and we do not select favourable seeds or checkpoints for any baseline.

E. Label dependence and cost

The pipeline fits and ranks without labels, but the number of segments and the peak budget still come from the annotations, and BIC could not replace the first. On cost, the dense VAE used comparable time and less memory than S^{3}PL on the section measured, and correlation ranking takes at most about a second, where the IG step, including its GMM fit, takes seconds (CAC) to minutes (GBM, RCC). The lightweight model is therefore practical, but it did not gain from its spatial input, which was the reason for adding it.

The conclusions are bounded to the tested models and datasets: we do not claim that spatial context never helps, that IG is generally inferior or that a nonlinear encoder is never needed.

F. Limitations

Each collection has eight sections, drawn from four PDX lines (GBM), one tumour (CAC) and eight subjects (RCC), so the statistics describe these sections rather than a wider population, and effect sizes carry most of the weight. Choosing the number of segments by BIC instead of the class count changed the GBM ranking result, and BIC chose the largest value tested, so a reliable label-free choice of K remains open. mSCF1 measures agreement with tissue masks rather than molecular identity, which favours methods that produce mask-like maps; the RCC masks include a large unannotated region, and the CAC masks are model predictions. Finally, we tested fixed-window neighbourhoods in a dense VAE. Graph-based and image-level spatial models, transfer between sections and other attribution settings remain open (section VI).

VI. Conclusion

This report tested whether local spatial context helps a lightweight VAE select informative MSI peaks and which part of a learned pipeline determines peak quality. Under matched budgets and predeclared rules, on three public collections, we reached the following conclusions.

- Ranking peaks by correlation with an unsupervised GMM segmentation outperformed gradient attribution through the VAE on every dataset when the number of segments matched the tissue classes; this was statistically supported (nominally) and in the same direction for every training seed. With the number of segments chosen without labels, it held on CAC and RCC but not on GBM.
- Neighbourhood input to the encoder and smoothing of the segmentation gave no consistent, spatially specific gain. Moran's I, which uses no labels, was the strongest label-free ranking on GBM and RCC and within 0.01 of the best on CAC.
- A linear PCA segmentation was not reliably worse than the VAE, and reconstruction error did not track peak quality.
- msiPL reproduced. S^{3}PL's published GBM score did not. Weight initialisation dominated seed sensitivity in two of three selected sections, while data order also mattered in one.

The contribution is a controlled evaluation, not a new peak picker. It shows that, in a learned MSI pipeline, the useful spatial signal lies in how peaks are ranked against a segmentation rather than in neighbourhood input to the encoder, and how strongly such conclusions depend on matched budgets, model-free references and seed replication.

Future work should first:

- validate a label-free choice of the number of segments, which BIC could not provide;
- test a predeclared combination of segmentation correlation and Moran's I on new data;
- evaluate spatial models that operate on graphs or whole ion images rather than fixed windows;
- test whether a model fitted on one section transfers to another.

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Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools [ ]

I **did** make use of generative AI tools for the following (tick all that apply):

1. Idea Generation (research problem/design, hypothesis) [X]
2. Sourcing Related Work (summarising, identifying sources) [X]
3. Methods and Experiment Design (experiment setup, model tuning) [X]
4. Data Analysis (presentation, coding, interpretation) [X]
5. Theoretical Development (theorem proving, conceptual analysis) [ ]
6. Code Development (generating algorithms, writing scripts) [X]
7. Presentation (rendering graphics, formatting) [X]
8. Editing (grammar, readability) [X]
9. Writing (text generation, document structuring) [X]
10. Citation Formatting (structuring, organising) [X]

If other uses were involved, please specify below:

Generative AI tool used (list all)	**Used for?**
ChatGPT (OpenAI)	Early supervision and planning: experiment ideas, decision rules and reviewing results.
Claude (Anthropic)	Supervision: planning experiments, decision rules, reviewing results against saved outputs, drafting and editing the report.
Claude Code (Anthropic)	Writing, testing and running analysis code and cluster jobs.
OpenAI Codex	Code review, commits and report editing and checking.

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number: 2653934

Candidate signature: 

Date: 7 November 2026

AI DECLARATION: HOW AI WAS USED, AND REPRESENTATIVE PROMPTS

Working system. I ran the project as a supervised workflow with three AI roles and a written, append-only supervision log. An AI supervisor (ChatGPT at first, then Claude) helped me plan experiments, proposed predeclared decision rules and checked each reported result against the saved outputs. Claude Code implemented and ran the code and cluster jobs and reported back in the log. The log was the only channel between the roles: each implementation task ran in a fresh Claude Code session briefed from the log, which kept each session focused on one task and limited usage, and the supervisor read and checked each report before the next step. OpenAI Codex independently reviewed every code change and, once I approved it, committed it under my account with messages I set rules for. The goals were fixed by me: a fair, matched-budget comparison; decision rules written before results; and no claim without a saved result behind it. Every experiment, cluster job, rule change and commit needed my explicit approval.

Writing. I had the assistant compile every verified result into an evidence dump organised by report section. I then chose what the report should contain, set its structure and framing, and had the assistant draft sections from that selection. I reviewed every draft, rejected or redirected large parts of it (for example the framing, the research questions, the figures and the statistical presentation) and approved the final text. The prompts below are a small selection from several hundred exchanges over the project, chosen to illustrate how methodological decisions and the writing were directed; they do not cover every use. They are quoted as written (typos corrected; [...] marks omitted text), and the full logs are available on request.

Questioning methods and results

- *ChatGPT:* "We only tested the neighbourhood's contribution to reconstruction but how did we know it wouldn't end up being better for the peak picking?"
- *ChatGPT:* "Does this mean that there's a chance that our model just doesn't perform better than a standard VAE?"
- *ChatGPT:* "How else do we properly test that spatial has an impact?"
- *ChatGPT:* "What about S3PL runtime? Are we finally fully grabbing that information to have proper and fair comparisons with our model?"
- *ChatGPT:* "didn't we only use 1 seed [...] do we need to run the experiments more times to verify?"
- *ChatGPT:* "I do not want to conclude that a neighbourhood has no contribution especially since we have a very simple neighbourhood that we are using [...] what would [our ablation study] look like?"
- *Claude:* "Why did we not end up using Mann–Whitney as a statistical analysis method?"
- *Claude:* "Why didn't we evaluate when the scores are no longer increasing and choose that as our epochs?" (I accepted the advice that this would tune on the evaluation labels; training loss is reported instead.)

Clarifying concepts

- *ChatGPT:* "Wait, S3PL is self-supervised while we are unsupervised, right?"
- *ChatGPT:* "What is first-layer L2? Is that S3PL?"

Reviewing my work and checking claims

- *Claude and Codex:* "check chat" (my standing instruction: the supervisor read the newest entries from the implementation sessions, verified them against the saved results and ruled before anything proceeded; Codex read them to review a change before committing it).
- *ChatGPT:* "[...] I have a progress.md with my thoughts and understandings. Can you review it [...] And also fact check it and let me know where my understanding is lacking."
- *Claude:* "I tried to attempt to replicate the baseline msiPL VAE. Is this sufficient, and can you help me with better evaluation similar to that in msiPL so I can know if I am on the right track?"
- *Claude:* "[...] I also found the CAC dataset with the masks [...] and this is what I have so far."
- *Claude:* "[...] I was wondering if you could review [my repository] as it is right now because I do not think my current results are good in comparison to what I want them to be."
- *ChatGPT:* "Quick, before I run the scripts: why are we using the patch size 9 and not 3 when 3 is the paper?"
- *ChatGPT:* "[...] it seems like those masks came from a model and not actually medically annotated so I do not know if I feel comfortable using that as a ground truth [...]"
- *ChatGPT:* "Can you review the new supervisor review that Claude has done for us. [...] Let me know whether or not you agree [...] before moving forward."
- *Claude:* "Check the S3PL paper for the links to the datasets and how we got them previously and add that to the dataset documentation."

Figures and diagrams

- *ChatGPT:* "Can you make me a diagram with how our model looks vs msiPL vs our proposed one vs S3PL?"
- *ChatGPT:* "[...] in the architecture figure the centre-only / uniform / attention part of the diagram is overlapping still [...]"

Documentation

- *ChatGPT:* "Can you also help me start documenting what we are doing in Obsidian, because as much as I like what we are doing I also want to understand what we are doing and why."
- *ChatGPT:* "Can you go through everything we did beginning to end and make me a bunch of md files? I want a main one to record the results, then one for the processes and the whys and methodologies [...]"
- *Claude:* "Make a note of everything [...] update the Notion."

Directing the report

- *Claude:* "Remove the proposal to outcome table [...] Write it like any of the papers we read."
- *Claude:* "Is RQ4 important for our paper so much so that it is a main question?"
- *Claude:* "Can we better phrase the helps, equivalent etc.? Maybe explain how we decide it and then use the terms after."